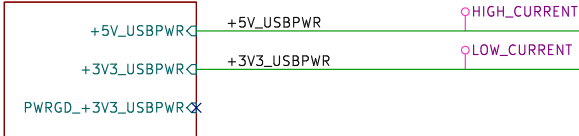


TODO:
- JTAG SHIFTERS POWERED FROM DATA USB

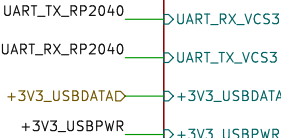
Future consideration:
- Current sense
<https://learn.adafruit.com/adafruit-ina219-current-sensor-breakout?view=all>

USB C #1 - Power Input, 3V3 Regulation and Fan Connector



File: usbc_power_in.kicad_sch

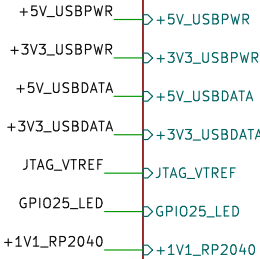
NOTE- UART SIGNAL LABELS CHANGE HERE



VCS3 UART and Power Connector

File: uart.kicad_sch

LEDs and Test Points



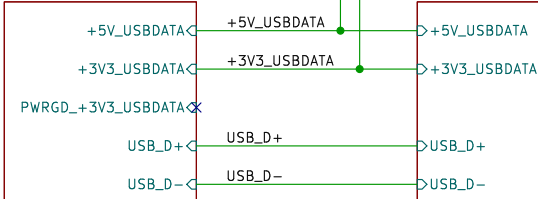
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MIPI Connectors



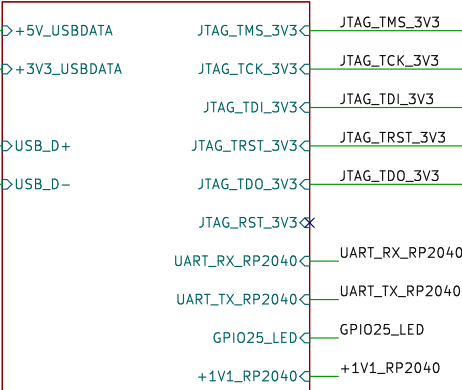
File: mipi_connectors.kicad_sch

USB C #2 - Data Input



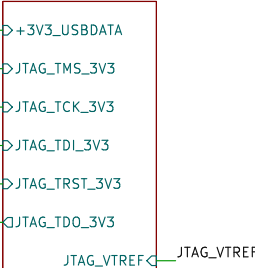
File: usbc_data_in.kicad_sch

RP2040 for JTAG and UART

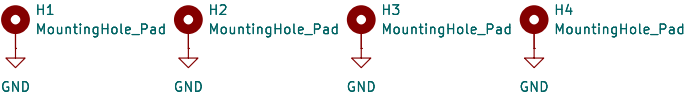
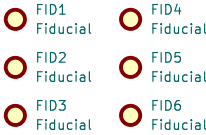


File: rp2040.kicad_sch

JTAG Level Shifter and Connectors



File: level_shifter.kicad_sch



Sheet: /
File: VCS RP2040 V3 SMALL.kicad_sch

Title: VCS3 Hub

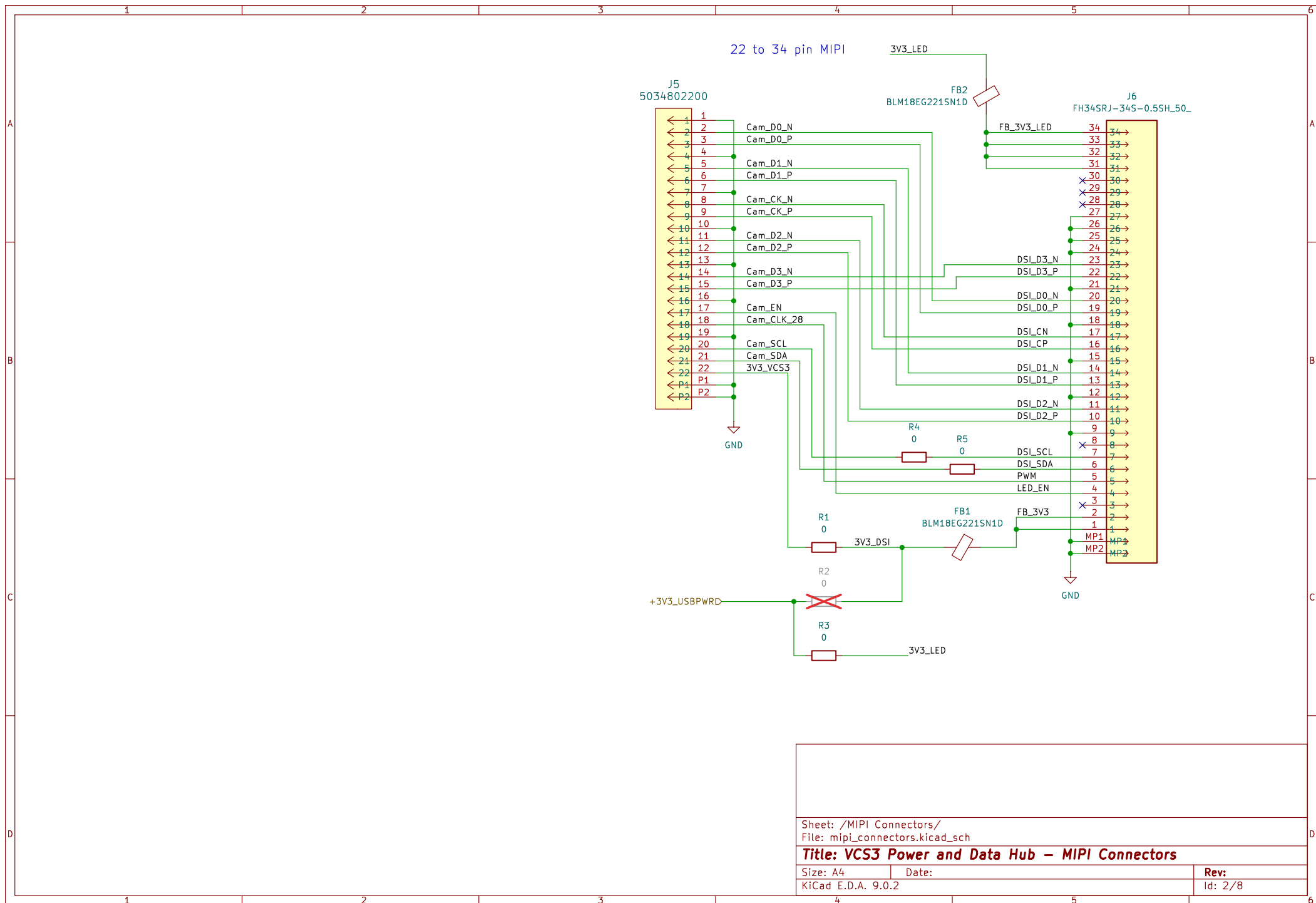
Size: A4

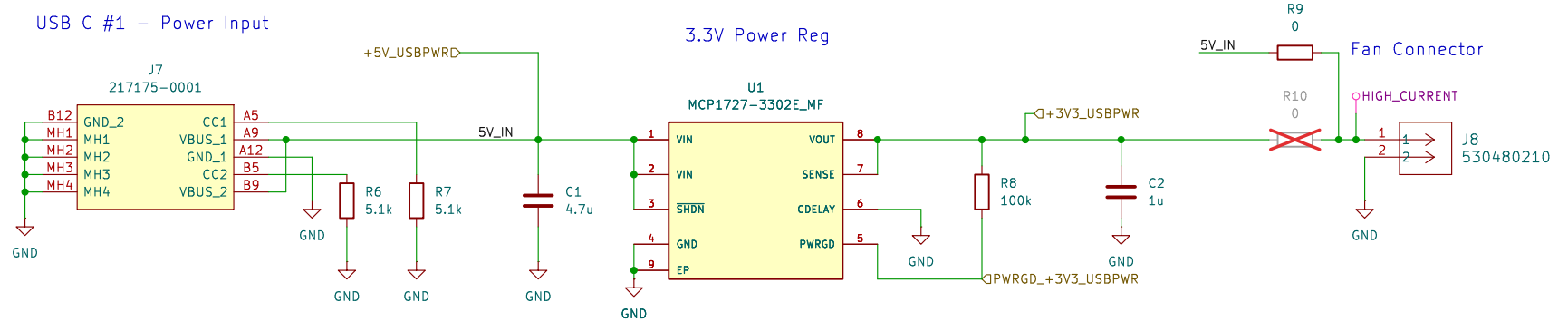
Date:

KiCad E.D.A. 9.0.2

Rev: 1

Id: 1/8





Sheet: /USB C #1 – Power Input, 3V3 Regulation and Fan Connector/
File: usbc_power_in.kicad_sch

Title: VCS3 Hub – Power Input

Size: A4
KiCad E.D.A. 9.0.2

Date:

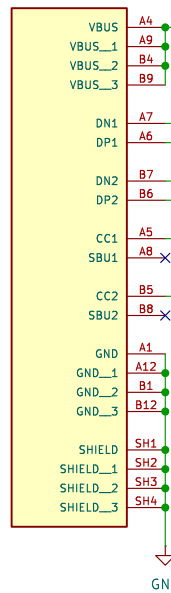
Rev:
Id: 3/8

USB C #2 - Data Input

J9
UJ20-C-H-G-TH-P16-TR

PWR_FLAG

+5V_USBDATA



+5V_USBDATA

R11
5.1k

R12
5.1k

PWR_FLAG

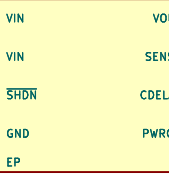
GND

C3
4.7u

GND

GND

U2
MCP1727-3302E_MF



R13
100k

C4
1u

GND

PWRGD_+3V3_USBDATA

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

Sheet: /USB C #2 - Data Input/
File: usbc_data_in.kicad_sch

Title: VCS3 Hub - Data Input

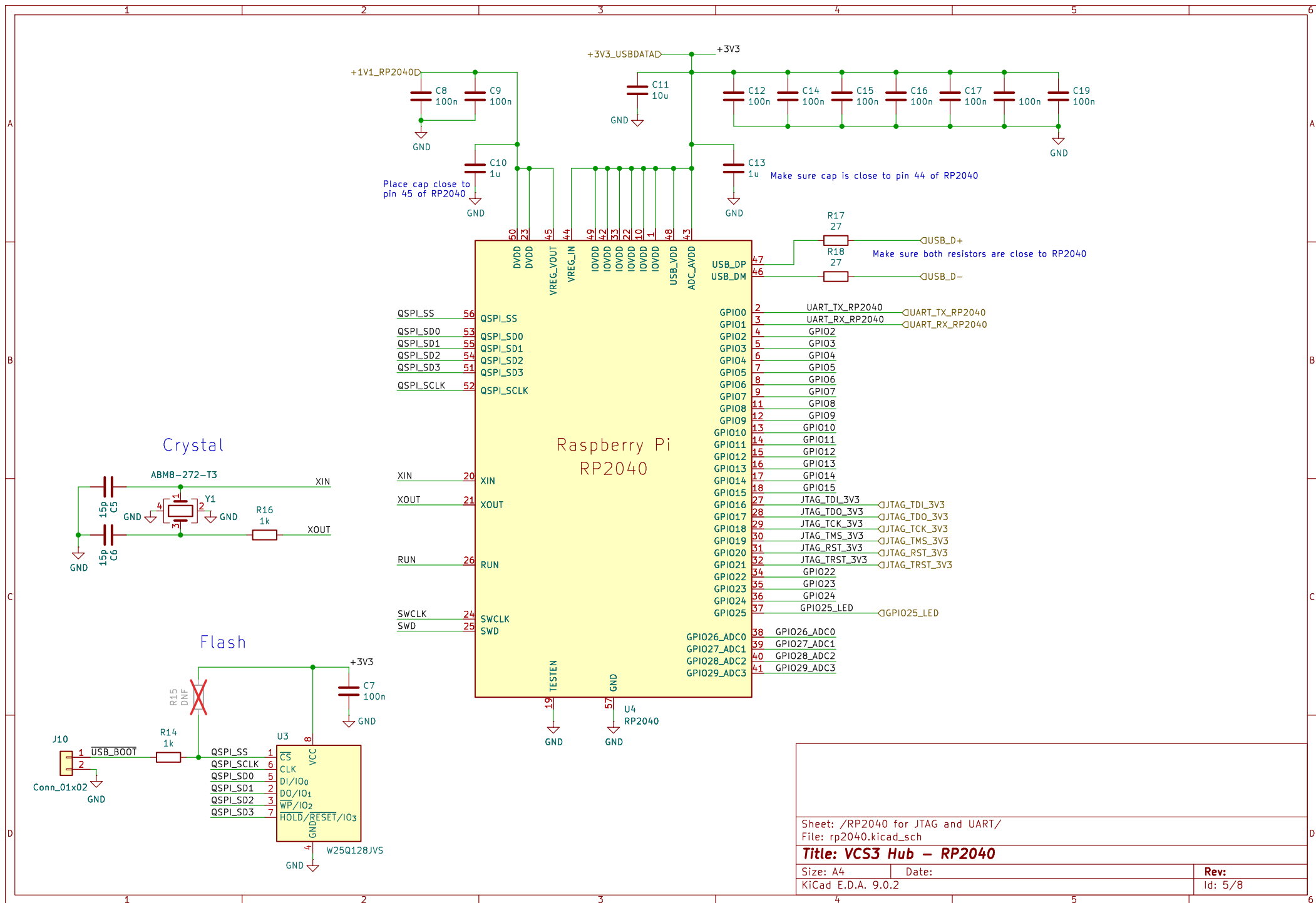
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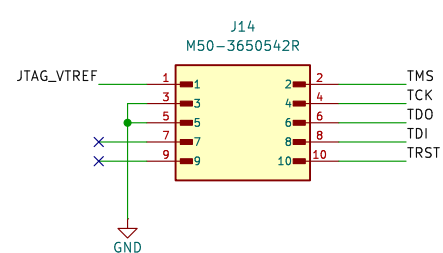
Date:

KiCad E.D.A. 9.0.2

Rev:

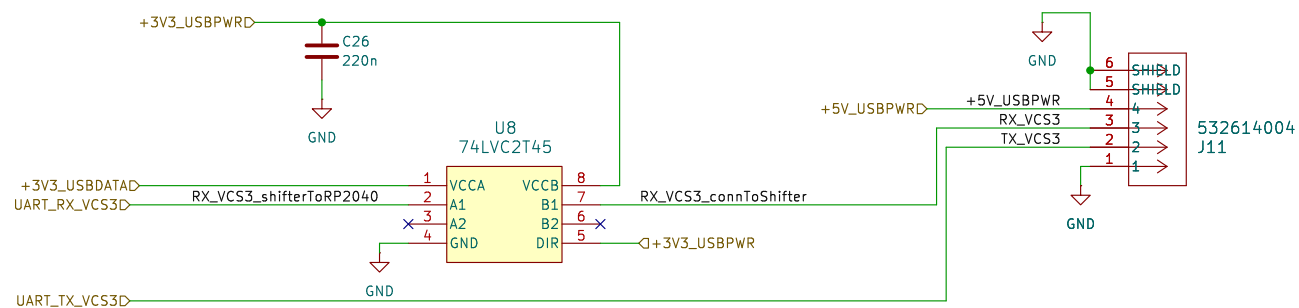
Id: 4/8





Id: 6/8

This level shifter is used as a buffer.
It prevents the "RX VCS3" signal from powering the VCS3 if there is no USB PWR connected.



Sheet: /VCS3 UART and Power Connector/
File: uart.kicad_sch

Title: VCS3 Hub – VCS3 UART and Power Connector

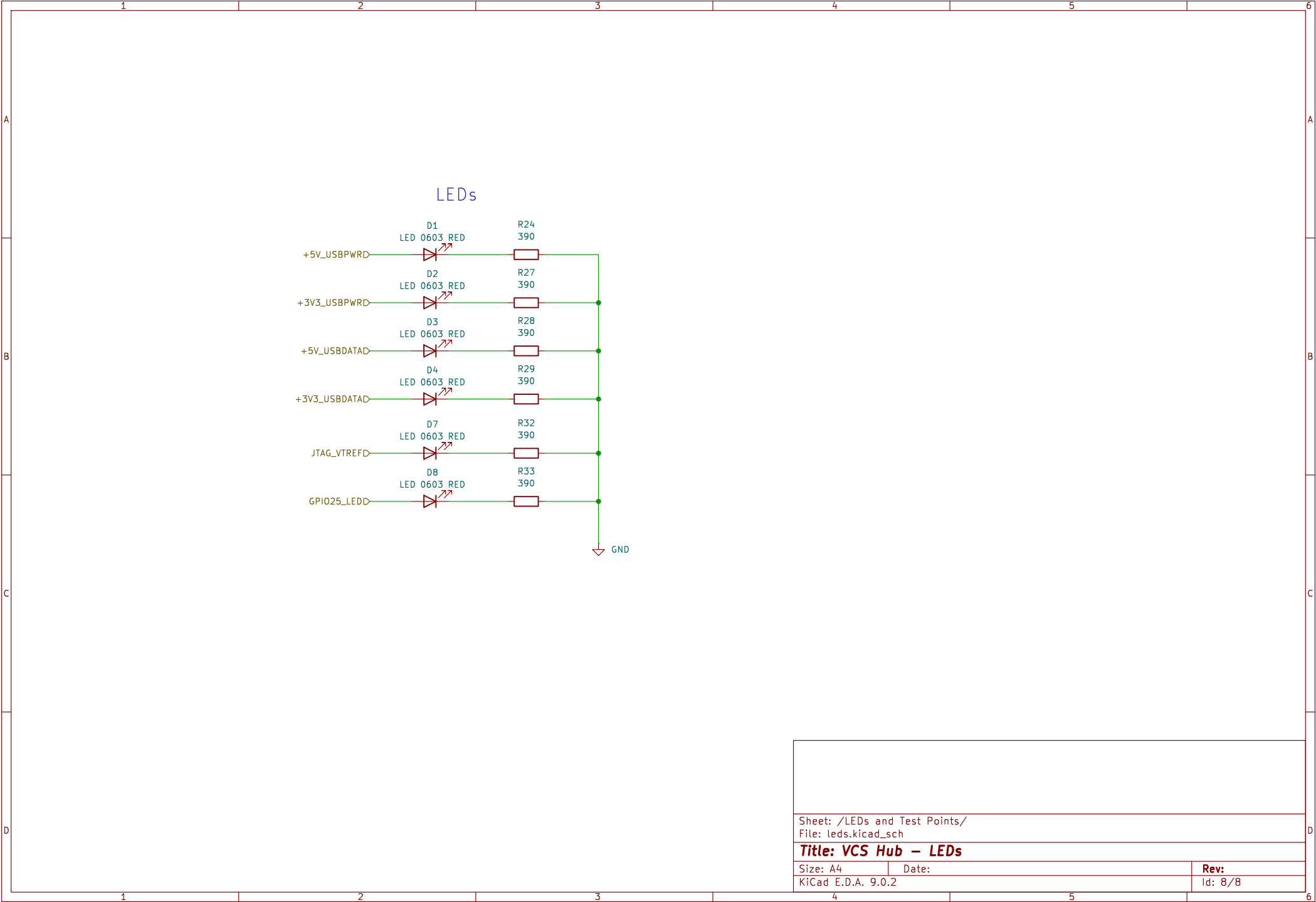
Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.2

Id: 7/8



Sheet: /LEDs and Test Points/ File: leds.kicad_sch		
Title: VCS Hub – LEDs		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.2	Id: 8/8	